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Chapter 01 – Introduction

1. Introduction

The modern academic environment for university students is increasingly reliant on digital tools for collaboration, note-sharing, and real-time communication. However, these tools are often fragmented, forcing students to navigate multiple platforms to manage their academic commitments. This disjointed approach not only interferes with productivity but also limits effective collaboration and knowledge exchange. Recognizing these challenges, our project proposes the development of an integrated Note-Sharing and Study Collaboration Platform.

This platform aims to centralize essential academic functionalities, including AI-powered learning support, real-time collaboration features, and an incentive-driven note-sharing system. By providing a unified digital workspace, the platform seeks to enhance student engagement, streamline study processes, and foster collaborative learning.

2. Definition of the Problem

University students face several critical challenges when it comes to managing their academic responsibilities efficiently.

The primary issues include:

- **Fragmented Digital Tools:** Students currently use a combination of platforms for different academic tasks — note-sharing apps, video conferencing tools, and AI chatbots for knowledge support. This fragmentation disrupts workflow and creates inefficiencies.
- **Lack of Real-Time Collaboration:** Existing note-sharing platforms often lack synchronous collaboration features. Students must resort to separate applications for real-time discussions, making it hard to work collectively on study materials.
- **Limited Incentives for Knowledge Sharing:** While some platforms allow students to share notes, they rarely offer recognition for contributors. This discourages the sharing of high-quality, value-added content.
- **Absence of Adaptive Learning Support:** Many collaboration platforms lack AI-powered functionalities to provide personalized learning experiences or instant academic assistance, leaving students without on-demand support.

These issues collectively hinder students' ability to learn efficiently, collaborate seamlessly, and engage with their academic work in a structured manner.

3. Project Objectives

The primary objective of this project is to design and develop a comprehensive Note-Sharing and Study Collaboration Platform tailored to the needs of university students. The specific goals of the project are as follows:

1. Integration of Academic Tools:

- Develop a unified platform that combines note-sharing, real-time collaboration (video conferencing, live chat), and AI-powered academic support.

2. Incentive-Based Knowledge Sharing:

- Implement a reward system (leaderboards, star ratings, reviews) to encourage students to share high-quality notes and study materials, similar to successful models in platforms like Course Hero.

3. Real-Time Collaboration Features:

- Integrate the Zoom API to enable seamless virtual study sessions within the platform.
- Facilitate group document editing and live discussions to promote collaborative learning.

4. AI-Driven Learning Support:

- Utilize AI models (like ChatGPT) to provide instant academic assistance, answering student queries and offering personalized study recommendations.

5. Scalable and Secure Architecture:

- Develop a robust backend using PHP, MongoDB databases.

4. Scope of the Project

The scope defines the boundaries and deliverables of the project. It outlines what the platform will and will not cover to ensure clear expectations.

In-Scope:

- Development of a centralized web platform integrating note-sharing, real-time collaboration, and AI-powered support.
- Implementation of an incentive-based system for content contributors.

- Integration of Zoom API for virtual study sessions and ChatGPT API for AI-driven learning.
- Designing a user feedback mechanism to gather continuous input.
- Establishing a secure and scalable backend system.

Out-of-Scope:

- Offline functionality or mobile application development (considered for future phases).
- Advanced tutoring services or direct educator involvement beyond note-sharing.
- Integration with third-party commercial educational platforms.

5. Summary

This Functional and Technical Specification report outlines the core aspects of the Note-Sharing and Study Collaboration Platform. The introduction has highlighted the fragmented nature of current academic tools and the need for an integrated solution.

The proposed platform will address these challenges by offering a seamless, AI-powered, and incentive-driven academic ecosystem for university students. With clearly defined objectives and a focused project scope, the platform aims to revolutionize collaborative learning, enhancing both student productivity and engagement.

Chapter 02 - System Analysis

1. Facts gathering techniques

Developing an online note-sharing platform requires a thorough understanding of user needs, system requirements, and potential challenges. To gather necessary data about effective system development, we used three key techniques which are observation, reviewing and brainstorming, this helps to identify user behaviours, pain points and common usage patterns, ensuring that the system meets user expectations.

- **Observations**

Based on our study, most people share notes in many ways. Most of the student prefer handwritten notes and share those notes physically in person and some share notes using messaging apps, emails and cloud storage. These sharing methods lacks content organization, quality control and specially limited accessibility.

- **Brainstorming and reviewing**

By monitoring different types of users and analysing existing note sharing platforms and methods to identify strengths and weaknesses. We conducted usability tests on existing platforms, like as Course Hero, Quizlet, Study drive and Oneclass to identify features, workflows,

data storage methods, UI/UX patterns and functionalities and also to distinguish potential drawbacks in those systems.

By combining system evaluation with the analysis of user behaviour, we are developing a note-sharing system that is instinctive, secure, as well as productive. We are developing a market differentiation with actual user needs and wants.

2. Current system

In present academic environment, Students exchange notes through various types of traditional and digital methods. One of the few traditional approaches is the use of photocopies or scanned copies of handwritten notes. Many students use this method because it allows them to share handwritten notes without an effort, especially when getting ready for upcoming exams, this method is often consuming lot of time and may lead to degradation of content quality, especially in hand written materials.

In additions to traditional methods, students often use digital means like messaging apps and emails to share notes in platforms like WhatsApp, Telegram and Messenger which allows direct exchange of text documents, images, PDF files almost immediately, however navigating through emails, text messages, or unorganized texts to find specific documents can lead to difficulties, particularly when managing multiple modules or subjects.

In these days most of the academicians utilizes cloud storage, where they can share notes using platforms such as Google Drive, One Drive, or Apple I cloud. These centralized services allow academicians to access and edit shared files, images and documents. These various types of digital and traditional methods helped students to find academic knowledge till the present time.

3. Overall Use Case Diagram for the Current System

3.1. Use case diagram

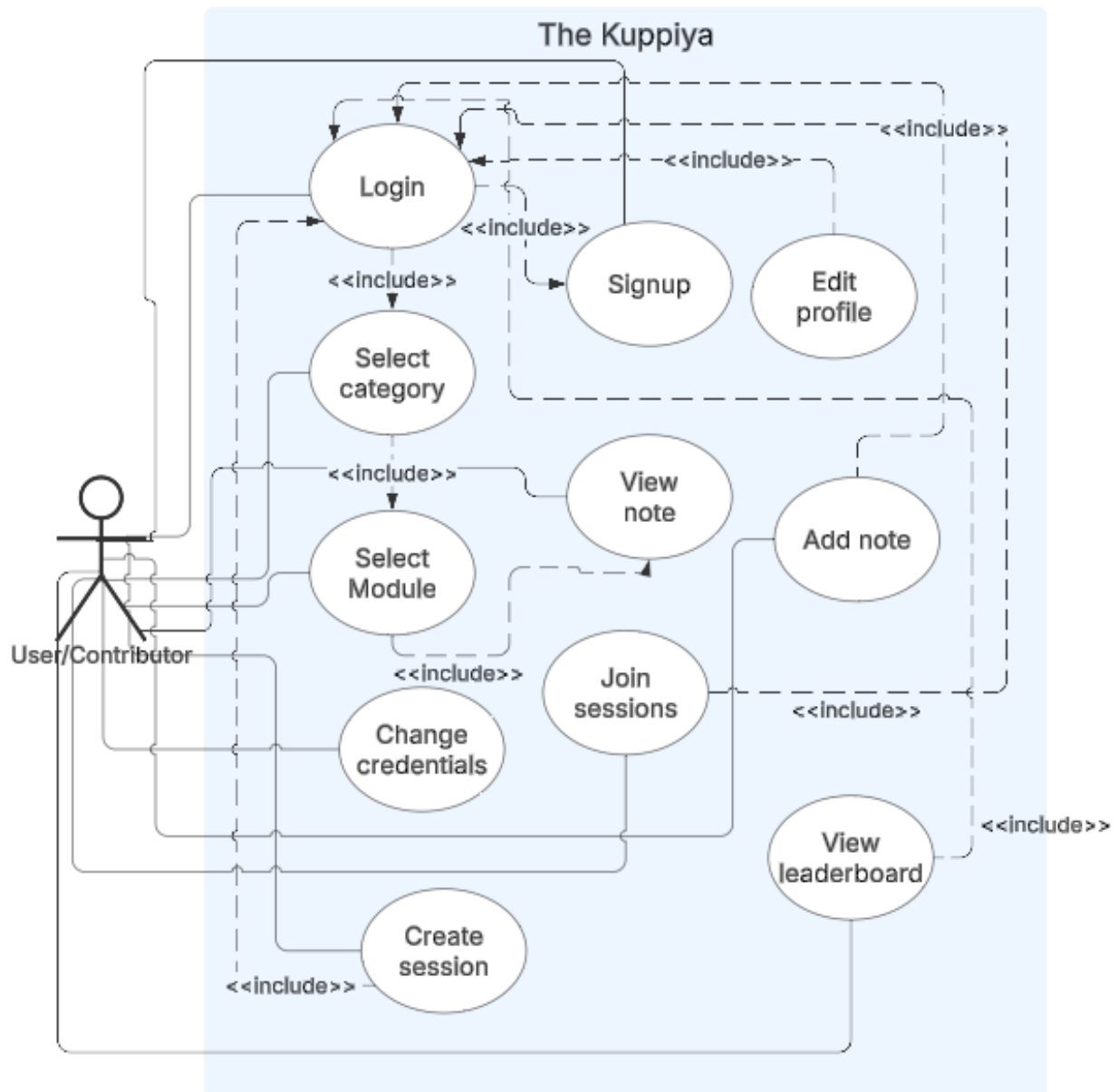


Figure 1: Use Case

3.2. Use case description

Use case	Use case description
Login	Allow User or contributor to authenticate credential details before accessing the system. This is required for all the use cases.
Signup	Allow user/contributor to register to the system by filling necessary credentials.
Select category	Allow user/contributor to select desired category.
Select module	Allow user/contributor to select desired module name after selecting category.
View notes	Allow users to browse and find necessary notes uploaded by contributors.
Add note	Allow contributors to add note/documents to relevant category and module.
Create session	Allow contributors to create sessions.
Join sessions	Allow users to join relevant sessions conducted by contributors.
Edit profile	Allow users to change their name, degree and university.
Change credentials	Allow users to change or delete their email and password.
View leaderboard	Allow users view most experience contributors in order.

Table 1

4. Drawbacks of the Current System

Despite the availability of multiple methods of sharing notes and documents, several drawbacks exist that affect the effectiveness of the student's learning experience. One of the major problems with the traditional methods is lack of accessibility. Physical notes easily be lost and damaged or can become unreadable due to poor quality printing. Additionally scanning handwritten notes can result in unclear images that may be difficult to interpret, especially when dealing with diagrams and mathematical equations. The amount of money needed for photocopying large number of notes can also be expensive, especially students with financial constraints.

Digital methods may offer suitable facilities, but also come with challenges. Social media apps for instance, are not designed for organized document sharing. Documents can get lost in long chat threads, making it hard to find relevant information when the student needed. And also, some instant messaging apps and platforms compresses image quality to save large number of data

exchange, which can lead to unclear and hard to interpret documents. Even though email specific sharing is structured, it becomes inefficient when dealing with large number of files as inboxes get cluttered with emails and searching specific notes will be time consuming.

Cloud based note sharing may be challenging when multiple users upload content without proper categorization. Another concern is that privacy and security, students may unintentionally share sensitive academic data or allow unauthorized access to their cloud.

Even though academicians have the ability to share their notes in multiple methods, each has its own drawbacks. The need of a more organized, interactive, and secure system is evident to improve accessibility, reliability, and efficiency in academic note sharing.

5. Summary

The present methods of notes sharing include both traditional and digital approaches. While these methods help students exchange study materials, they also have some issues like poor content organization, quality degradation, accessibility, limitations and security concerns. To develop more efficient and effective online note sharing platform, we observed current methods, brainstorm and review efficient system evaluation techniques.

The use case diagram illustrates how users interact with the new system, with functionalities like user authentication, browsing of notes, uploading, categorization, session creation and leaderboard ranking.

Despite the availability of various kind of methods, draw backs exists. Traditional note sharing methods suffer from accessibility and quality issues, while digital approaches struggle with categorization, security and reliability. These challenges highlight the need of more efficient, interactive and secure note sharing platform enhances accessibility, categorization and collaboration for academicians.

Chapter 03 - Requirement Specification

1. Functional Requirements

The system should provide the following functionalities:

1.1. User Authentication & Profile Management

- Users must sign up with an email and password.
- Users must provide additional details: name, gender, university, graduation date, and study program.

- Users can log in and log out securely.
- Users can view and edit their profile details.

1.2. Notes Management

- Users can upload study notes with the following details:
 - Main Category (e.g., IT, Business, Science).
 - Subcategory (e.g., Programming, Databases).
 - Language of the note.
 - Document Type (PDF, DOCX, PPT).
 - Description of the note.
- Users can view notes by filtering Main Category → Subcategory.
- Users can rate notes (1-5 stars).
- Users can comment on notes for discussion.

1.3. Discussion Section

- Each subcategory has a discussion section where users can:
 - Create a post (text or image).
 - Like posts from other users.
 - Comment on posts to interact.

1.4. Online Sessions (via Zoom API)

- Users can create learning sessions by providing:
 - Topic
 - Hosted By
 - Date & Time
 - Duration
- Users can join sessions by filtering categories or from the recommended list.

1.5. Leaderboard & XP System

- Users earn XP points for:
 - Uploading notes.
 - Viewing and rating notes.
 - Creating and joining sessions.
 - Participating in discussions (posting, liking, commenting).
- The leaderboard ranks users based on XP, with the highest XP user at the top.

2. Non-Functional Requirements

These requirements define the overall system behavior:

- **Usability:** The system should have an intuitive UI for easy navigation.
- **Scalability:** It should support a growing number of users and content.
- **Maintainability:** The system should allow easy updates and bug fixes.
- **Compatibility:** It should work on desktop and mobile browsers.
- **Availability:** The website should be accessible 24/7.

3. Performance Requirements

- The system should load pages within 3 seconds.
- The database should handle at least 1,000 concurrent users.
- Uploaded notes should be retrievable within 2 seconds.
- The system should process user actions (e.g., likes, comments, XP updates) without noticeable delay.

4. Security Requirements

- **User Authentication:** Secure login system with password encryption.
- **Data Protection:** Prevent unauthorized access to user data and notes.
- **Secure File Uploads:** Validate and scan uploaded documents for threats.
- **Session Security:** Implement token-based authentication for secure access.
- **Prevent Spam & Abuse:** Restrict excessive note uploads or comments from a single user in a short time.

5. Hardware Requirements

Client-Side (Users' Devices):

- Any device with a modern web browser (PC, laptop, tablet, smartphone).

Server-Side (Hosting Requirements):

- Web Server (e.g., Apache/Nginx).
- Database Server (e.g., MongoDB).
- At least 8GB RAM and 100GB storage for smooth operation.

6. Networking Requirements (Optional)

- Minimum 10 Mbps internet speed for smooth Zoom sessions.
- Cloud hosting with 99.9% uptime for availability.

7. Safety Requirements (Optional)

- Regular backups to prevent data loss.
- Server should have firewall protection against cyberattacks.

8. Summary

This chapter defined the functional and non-functional requirements of **The Kuppiya**. The system will provide a note-sharing platform, discussion section, and online sessions with an XP-based leaderboard. It must be secure, scalable, and user-friendly while ensuring high performance and availability.

Chapter 04 - Feasibility Report for the Study Collaboration and Note-Sharing Platform

1. Financial Feasibility

For the project to be sustainable in the long run, it must be economically viable, ensuring that the costs of development and maintenance are balanced by potential revenue streams.

Key Costs Involved

- **Cloud Infrastructure:** Since the platform will be hosted on cloud services like AWS, Google Cloud, or Microsoft Azure, monthly costs will include server maintenance, data storage, and bandwidth usage to handle user activity efficiently.
- **API Subscriptions:** The platform integrates third-party APIs like Zoom (for video conferencing) and ChatGPT / Deepseek API (for AI-powered assistance). These APIs usually require subscription fees based on usage.

Revenue Generation Strategies

To cover expenses and ensure long-term sustainability, the platform can generate revenue through:

- **Advertising:** Partnering with educational institutions, publishing companies, or online course providers can allow relevant in-platform advertisements that benefit both the students and the advertisers.
- **Sponsorship Opportunities:** The platform can seek financial support from universities, government education programs, or learning technology companies that wish to promote academic collaboration and knowledge-sharing.

A detailed financial analysis suggests that with strategic pricing and effective monetization, the platform has the potential to be profitable while remaining accessible to students.

2. Operational Feasibility

Operational feasibility assesses whether the platform can function efficiently and effectively while meeting the academic needs of students.

Key Success Factors

- **User Acceptance:** Since the platform is designed for students, its success depends on whether it meets their expectations. Surveys and beta testing will be conducted to gather feedback and refine features accordingly.

- **Platform Efficiency:** The system should handle real-time collaboration / video conferencing, and AI-based assistance without delays or crashes. Performance optimization will be a priority.
- **Scalability:** The platform must be designed to accommodate growth—it should support thousands of concurrent users without affecting speed, data retrieval, or overall stability. Cloud hosting solutions allow automatic scaling to handle increased user activity.

The goal is to ensure that students can seamlessly interact, share notes, join discussions, and receive AI-powered learning support without any technical difficulties.

3. Technical Feasibility

This section evaluates whether the chosen technology stack and infrastructure can support the platform’s functionality and ensure a smooth user experience.

Technology Stack

- **Frontend Development:** The platform will use Bootstrap libraries to create an interactive, responsive, and user-friendly interface that works across different devices.
- **Backend Development:** PHP will be used to handle data processing, authentication, and system logic, ensuring fast and secure interactions between users and the database.
- **Database Management:** MongoDB will store user profiles, notes, discussions, and rewards while ensuring efficient data retrieval and security.
- **Video Conferencing:** The Zoom API will be integrated to allow students to host and join live study sessions directly within the platform.
- **AI Support:** The ChatGPT API / Deepseek API will provide personalized learning assistance by answering student queries and offering study recommendations.
- **Data Security:** Authentication methods and encryption techniques will be implemented to ensure data privacy and secure user interactions.

By using modern web technologies, the platform will be scalable, efficient, and secure, ensuring a high-quality learning experience for students.

4. Organizational Feasibility

Organizational feasibility assesses whether the development team has the necessary skills and resources to build and maintain the platform effectively.

Key Organizational Aspects

- **Agile Development Methodology:** The team will follow an Agile approach, which means iterative development, continuous testing, and regular user feedback will shape the platform's evolution.
- **Team Structure:** The development team consists of software engineering students with expertise in frontend and backend development, database management, API integration, and UI/UX design. Clearly defined roles will help in efficient task management and collaboration.
- **Post-Launch Support:** After launching the platform, the team will focus on bug fixes, feature updates, and security enhancements to ensure the platform remains reliable and up to date.

By ensuring structured teamwork, clear communication, and ongoing maintenance, the project will remain operationally and technically sustainable.

5. Outline Budget

	Items	Estimated Cost (LKR)	Remarks
APIs	Zoom API	0	Free plan supports basic meeting capabilities
	Open AI (ChatGPT API) or other	6500	Limited usage for testing and integration
Frontend Development	UI/UX design	0	Used free software like Figma
	Frontend and frameworks	0	Used free and open-source software
Database	MongoDB Atlas	0	Free tier for the project
Backend Development	PHP Development language	0	Customizing own code for the project
Domain & hosting	Free subdomain E.g. GitHub pages, Vencel	0	Using free domains
	Hosting	1500	Using AWS free tier or Heroku

Other cost	SSL certificate	0	Use free security enhancements
	Third-party integrations	2000	Use cost effective software tools
	Frame works	0	Using free and open-source frameworks
Total Cost		10 000	Concerning on budget friendly deployment

Table 2

6. Summary

The feasibility report confirms that the Note-Sharing and Study Collaboration Platform is viable in terms of economic, operational, technical, and organizational feasibility.

- Financially, the platform has a well-defined cost structure and revenue model, ensuring sustainability through freemium services, advertisements, and sponsorships.
- Operationally, it focuses on user needs, efficiency, and scalability, ensuring smooth performance even with a large user base.
- Technically, it leverages modern technologies like MongoDB and AI APIs, ensuring high performance, security, and real-time collaboration.
- Organizationally, the project team has the necessary expertise, follows an Agile approach, and has a structured plan for development and post-launch support.

By integrating note-sharing, AI-powered learning, and real-time study collaboration into a single efficient and scalable platform, this project has high potential for success.

Chapter 05 – System Architecture

1. Class Diagram

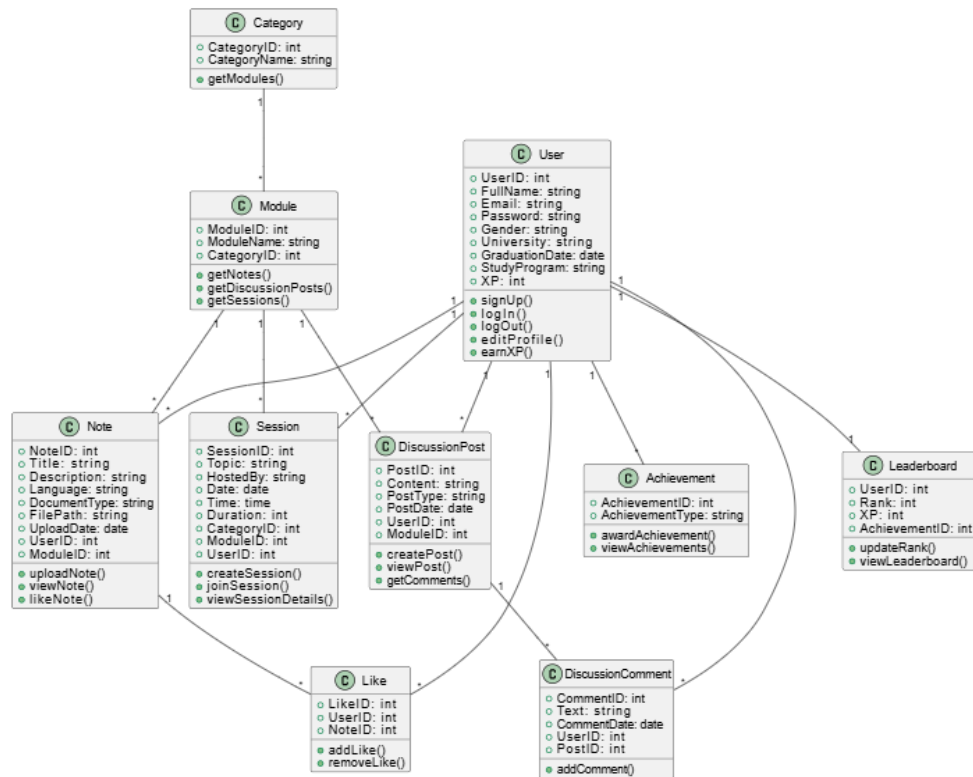


Figure 2 : Class Diagram

2. ER Diagram

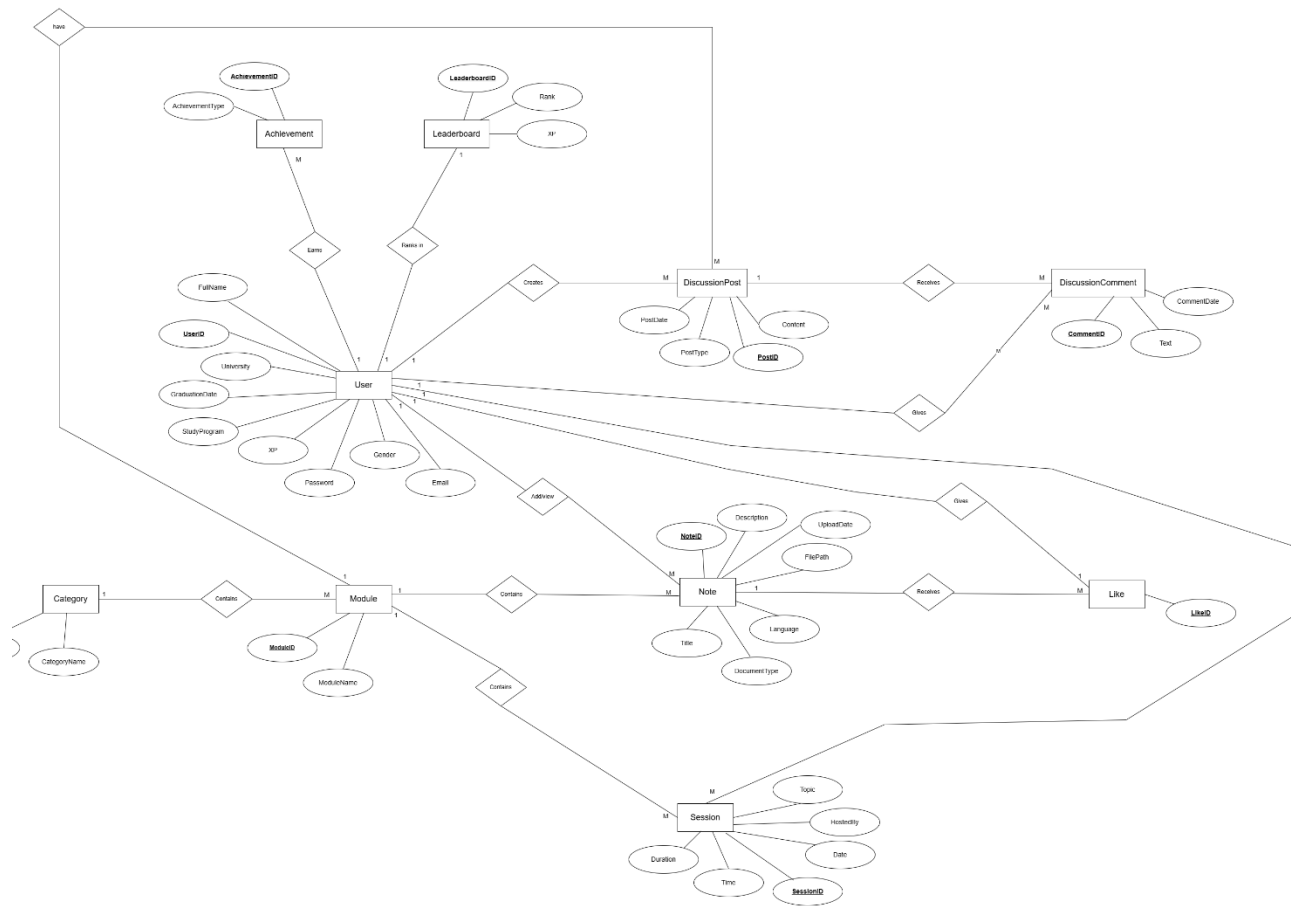


Figure 3 : ER Diagram

3. High-Level Architecture Diagram

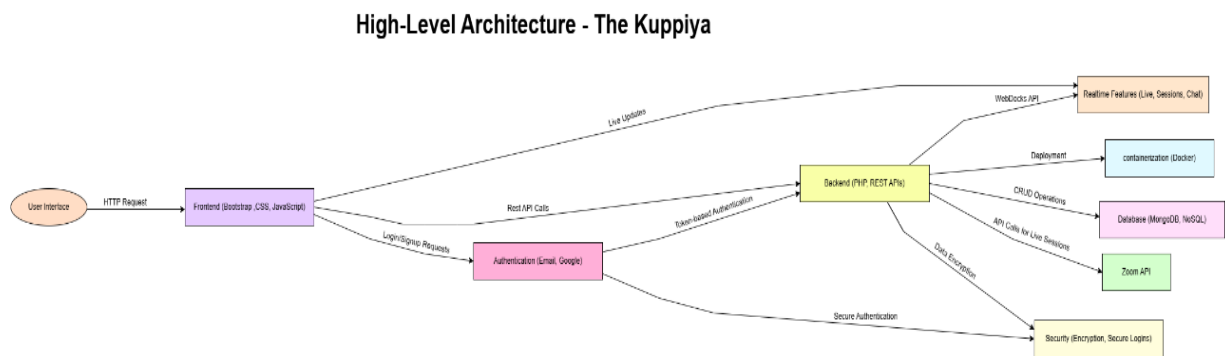


Figure 4 : High Level Architecture Diagram

Chapter 06 – Development Tools and Technology

1. Development Methodology

The platform follows an Agile development methodology, which allows for iterative development and continuous improvement based on user feedback. The team will focus on user-centered design, meaning that student needs will drive the development process. Surveys will be conducted to understand the common challenges students face when using multiple study tools. Based on this research, the platform will be built in stages, with regular testing and refinements to ensure it effectively addresses these issues. This approach ensures that the final product remains relevant, efficient, and user-friendly.

2. Programming Languages and Tools

To build the platform, the following technologies will be used:

- **Frontend (User Interface):** Developed using HTML, CSS, and JavaScript, ensuring a modern, responsive, and user-friendly design.
- **Backend (Server & Logic):** Built using PHP which is a powerful frameworks for handling data, managing user authentication, and processing requests efficiently.
- **Database (Data Storage):** The platform will use MongoDB to store notes, user profiles, and collaboration data.

3. Third-Party Components and Libraries

To enhance the platform's capabilities, various third-party components and APIs will be integrated. The ChatGPT API will provide AI-powered learning assistance, helping students receive instant answers to their queries. For real-time study sessions, the Zoom API will be incorporated, allowing students to connect through video conferencing directly within the platform. Additionally, Selenium and Jest will be used for testing, ensuring that the system is stable and free from major bugs. Version control will be managed through GitHub, allowing multiple developers to collaborate effectively and track changes throughout the project lifecycle.

4. Hardware Tools

The platform will be hosted on cloud services such as AWS, Google Cloud, or Azure to ensure high availability, security, and scalability. Cloud servers will handle data storage, AI computations, and video conferencing, optimizing the platform's performance. By leveraging cloud infrastructure, the system will be able to support a large number of users simultaneously, ensuring a smooth and uninterrupted learning experience.

5. Overall Summary

The Note-Sharing and Study Collaboration Platform is designed to address the inefficiencies students face when using multiple, disconnected platforms for academic purposes. Currently, students rely on separate applications for note-sharing, study discussions, and real-time collaboration, which leads to disorganization and a fragmented learning experience. This project aims to create a centralized, interactive, and AI-powered platform that integrates note-sharing, study groups, AI-driven learning support, and real-time discussions into a single, user-friendly system.

The platform's key objectives include enabling students to share and access high-quality study materials, participate in live study sessions, and receive AI-powered assistance for academic queries. To encourage participation, a reward system with leaderboards, stars, and ratings will be implemented, ensuring that users are motivated to contribute valuable content. The platform will also feature Zoom API integration, allowing students to host and join live discussions without needing external tools.

To develop this system efficiently, the project follows an Agile development methodology, ensuring iterative improvements based on user feedback. The technology stack includes React.js for the frontend, Node.js or Django for the backend, and PostgreSQL or MongoDB for data management. Real-time collaboration will be achieved using WebSocket technology, while AI-powered features will be integrated using ChatGPT API. The system will be hosted on cloud services such as AWS, Google Cloud, or Azure to ensure scalability and reliability.

The feasibility study confirms that the project is technically and economically viable. The economic feasibility accounts for cloud hosting, API licensing, and development costs, while the operational feasibility ensures that students can easily adopt and use the platform. Technical feasibility has been evaluated through the selection of a robust and scalable technology stack, ensuring high performance and security.

The platform's architecture includes a well-structured database design to manage users, notes, discussions, and real-time collaboration data. Algorithms such as AI-driven recommendations for ranking notes and real-time messaging through WebSockets will enhance the overall user experience. The system will also feature strong security measures, including user authentication, data encryption, and secure API connections, to protect user data.

By combining note-sharing, AI-powered assistance, and live collaboration into a single all-in-one academic platform, this project aims to revolutionize the way students learn, collaborate, and engage in knowledge-sharing. With its modern technology stack, user-friendly interface, and incentive-driven model, the platform will provide a seamless and interactive learning experience, making studying more efficient and engaging for students.

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Team Plan and Responsibilities

Member	Plymouth index no.	Name	Tasks carried out	Future plan	Comments by the group leader
1	10952961	Walimuni Rahul Mendis Abeysekara Wijayakulathilaka (Leader)	Leaderboard UI Design, Frontend Coding, XP Calculation Logic	Backend integration, Implement ranking logic, Improve XP tracking	I played the Part well.
2	10952979	Sunilge Siripadma Ruchira Ilangasinha	Sessions Frontend, Frontend Coding, Session Filtering	Improve session management, Implement Zoom API integration	The session UI is well-designed and works efficiently.
3	10952772	Balasinghege Sarali Thiloka	Dashboard UI Design, Frontend Coding, Display Metrics (Total Notes, Sessions, Users)	Enhance Dashboard UI, Integrate real-time updates	Clean and structured dashboard with clear metrics.
4	10952987	Mathara Arachchige Malith Ashen Chularathana	Add Note UI Design (Note Submission Form), Frontend Coding, Note Filtering Implementation	Backend integration, Implement rating and commenting system	The note submission feature is smooth and user-friendly.

5	10952728	Mahara Mudiyansele Dulni Thimethya Sandasini	User Profile UI Design, Edit Profile UI Design, Landing Page UI Design, Profile Viewing Feature	Implement Profile Picture Upload, Improve UI/UX of Profile Page	Well-organized profile UI with good navigation.
6	10953025	Gimhan Pramodya Algama	Signup/Registration Form UI Design, Settings UI Design, User Authentication UI	Backend Integration with MongoDB, Implement Email Verification	Registration form UI is simple and effective.
7	10952752	Uwage Don Nethum	Create Session / Join Session UI Design, Signup/Registration Form Frontend, Session Details Page	Backend integration, Implement session reminders and notifications	The session creation process is intuitive and well-structured.
8	10952855	Ravith Sandil Samarasekara	Settings Frontend, Account Preferences UI	Improve settings customization options	Settings page is well-designed and easy to navigate.
9	10952725	Manchanayaka Arachchige Doni Thisun	Login UI Design, Login Frontend, User Authentication UI Flow	Backend authentication, Forgot Password Feature	The login page has a clean layout and functions well.
10	10952794	Agampodi Kamodya Sandaranga	Landing Page Frontend, Introduction UI, Website Navigation	Improve landing page responsiveness, Implement animation effects	The landing page is visually appealing and well-structured.

Table 3